

Rebecca has \$1.90 worth of nickels and dimes. She has 2 more nickels than dimes.

1. Define the variables to write the system of equations.

Let x = number of nickels

Let y = number of dimes

2. Write a system of equations to determine the number of nickels and dimes Rebecca has.

$$\begin{cases} 0.05x + 0.10y = 1.90 \\ x - 2 = y \end{cases}$$

3. Solve the system using the method of your choice.

$$\begin{aligned} 0.05x + 0.10(x - 2) &= 1.90 & y &= (14) - 2 \\ 0.05x + 0.1x - 0.2 &= 1.90 & y &= 12 \\ 0.15x &= 2.10 \\ x &= 14 \end{aligned}$$

4. Rebecca has 14 nickels and 12 dimes.

Sampson and Riki went to the movies to watch 'The Batman' that was just released. Sampson bought 2 bags of popcorn and 5 pretzels, costing a total of \$37.25. Riki bought 8 bags of popcorn and 9 pretzels, costing a total of \$102.25. Determine the price of each bag of popcorn and the price of each pretzel.

5. Sampson created the equation $2b + 5p = 37.25$. Create a second equation that could be used to find the price of each bag of popcorn and the price of each pretzel.

$$8b + 9p = 102.25$$

6. Solve the system using the method of your choice.

$$(2b + 5p = 37.25) \cdot 4 \rightarrow 8b + 20p = 149.00$$

$$\begin{aligned} 8b + 20p &= 149.00 \\ -8b - 9p &= -102.25 \\ \hline 11p &= 46.75 \\ p &= 4.25 \end{aligned}$$

$$\begin{aligned} 2b + 5(4.25) &= 37.25 \\ 2b + 21.25 &= 37.25 \\ 2b &= 16 \\ b &= 8 \end{aligned}$$

7. The price of each bag of popcorn is \$8. The price of each pretzel is \$4.25.

A new Marvel movie was released at the Imax theater. The Friday 8 P.M. showing sold a total of 150 tickets. A child ticket costs \$7.75 and an adult ticket costs \$10.25. The show generated a total of \$1,470.

8. Write a system of equations that could be used to find the number of child tickets that were sold and the number of adult tickets that were sold.

$$\begin{cases} x + y = 150 \\ 7.75x + 10.25y = 1,470 \end{cases}$$

9. Solve the system of equations you wrote.

$$x + y = 150 \rightarrow y = 150 - x$$

$$7.75x + 10.25(150 - x) = 1,470$$

$$7.75x + 1,537.5 - 10.25x = 1,470$$

$$-2.5x = -67.5$$

$$x = 27$$

$$y = 150 - (27)$$

$$y = 123$$

27 children tickets and 123 adult tickets were sold.

Dale wants to build a new deck for his house. The nails come in large boxes and small boxes. The large boxes each have 25 nails, and the small boxes hold 20 nails. Dale bought a total of 295 nails and a total of 13 boxes.

10. Write a system of equations to find how many boxes of small and large boxes of nails Dale bought.

$$\begin{cases} 25l + 20s = 295 \\ l + s = 13 \end{cases}$$

11. Solve the system of equations you wrote.

$$l + s = 13 \rightarrow l = 13 - s$$

$$25(13 - s) + 20s = 295$$

$$325 - 25s + 20s = 295$$

$$-5s = -30$$

$$s = 6$$

$$l = 13 - (6)$$

$$l = 7$$

Dale bought 7 large boxes and 6 small boxes of nails.